

Code:

```
#include <stdio.h>
#include <stdlib.h>

#define MAX_FRAMES 3

void printFrames(int frames[], int n)
{
    for (int i = 0; i < n; i++)
    {
        if (frames[i] == -1)
        {
            printf(" - ");
        }
        else
        {
            printf(" %d ", frames[i]);
        }
    }
    printf("\n");
}

int main()
{
    int frames[MAX_FRAMES];
    int usageHistory[MAX_FRAMES]; // To store usage history

    // Initialize frames and usage history
    for (int i = 0; i < MAX_FRAMES; i++)
    {
        frames[i] = -1;
        usageHistory[i] = 0;
    }

    int pageFaults = 0;

    int referenceString[] = {7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2};
    int n = sizeof(referenceString) / sizeof(referenceString[0]);

    printf("Reference String: ");
    for (int i = 0; i < n; i++)
    {
        printf("%d ", referenceString[i]);
    }
    printf("\n\n");

    printf("Page Replacement Order:\n");

    for (int i = 0; i < n; i++)
    {
        int page = referenceString[i];
        int pageFound = 0;

        // Check if page is already in memory
        for (int j = 0; j < MAX_FRAMES; j++)
        {
            if (frames[j] == page)
            {
                pageFound = 1;

                // Update usage history
                for (int k = 0; k < MAX_FRAMES; k++)
                {
                    if (k != j)
                    {
                        usageHistory[k]++;
                    }
                }

                usageHistory[j] = 0;
                break;
            }
        }

        if (!pageFound)
        {
            printf("Page %d -> ", page);

            // Find LRU page (max usageHistory)
            int lruPage = 0;
            for (int j = 1; j < MAX_FRAMES; j++)
            {
                if (usageHistory[j] > usageHistory[lruPage])
                {
                    lruPage = j;
                }
            }
            int replacedPage = frames[lruPage];
            frames[lruPage] = page;
            usageHistory[lruPage] = 0;

            if (replacedPage != -1)
            {
                printf("Replace %d with %d: ", replacedPage, page);
            }
            else
            {
                printf("Load into an empty frame: ");
            }
        }

        printFrames(frames, MAX_FRAMES);
        pageFaults++;
    }

    printf("\nTotal Page Faults: %d\n", pageFaults);

    return 0;
}
```

Output:

```
C:\Users\Dell\Downloads\Unt x + v
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2
Page Replacement Order:
Page 7 -> Load into an empty frame: 7 - -
Page 0 -> Replace 7 with 0: 0 - -
Page 1 -> Replace 0 with 1: 1 - -
Page 2 -> Replace 1 with 2: 2 - -
Page 0 -> Replace 2 with 0: 0 - -
Page 3 -> Replace 0 with 3: 3 - -
Page 0 -> Replace 3 with 0: 0 - -
Page 4 -> Replace 0 with 4: 4 - -
Page 2 -> Replace 4 with 2: 2 - -
Page 3 -> Replace 2 with 3: 3 - -
Page 0 -> Replace 3 with 0: 0 - -
Page 3 -> Replace 0 with 3: 3 - -
Page 2 -> Replace 3 with 2: 2 - -

Total Page Faults: 13

Process returned 0 (0x0) execution time : 0.945 s
Press any key to continue.
```